From Theory to Practice: A Multi-Agent LLM System for Context-Aware Resume Analysis and Intelligent Job Matching

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This paper presents the development and evaluation of a production-ready multi-agent system for resume analysis that bridges academic research with practical implementation. Building upon recent advances in Large Language Model (LLM) architectures, particularly multi-agent frameworks and Retrieval-Augmented Generation (RAG), this work demonstrates how reasoning models like Llama 3.3 70B enable capabilities fundamentally beyond traditional embedding-based approaches. Through extensive testing with both synthetic and authentic resumes, we validate key architectural decisions including modular agent design, human-in-the-loop validation, and ten-dimensional career analysis. Our findings reveal that authentic resume data quality dramatically impacts system effectiveness, that geographic-aware intelligence is essential for actionable recommendations, and that the shift from encoding models to reasoning models represents a qualitative leap in career guidance capabilities. The system achieves strong alignment with human career counseling while processing resumes in seconds, demonstrating that advanced LLMs can deliver personalized, explainable career insights at scale.

Key Words: multi-agent systems, LLMs, resume analysis, career guidance, RAG, explainable AI, job matching, Llama 3.3 70B, Groq

1 Introduction

The recruitment landscape faces a fundamental paradox: while job seekers need personalized guidance to navigate increasingly complex career paths, traditional systems optimize for recruiter efficiency rather than candidate success. Recent research demonstrates that AI-driven resume screening can achieve correlation coefficients up to 0.84 with human evaluators [2], yet most systems remain opaque, offering scores without explanations or categorization without insight.

This work addresses a critical gap identified through both academic research and practical implementation: the need for AI systems that serve job seekers directly rather than merely automating recruiter workflows. While existing approaches focus on resume clustering for batch processing [1] or generation authenticity [2], we present a holistic career guidance system that analyzes, advises, and enables action.

The central thesis of this paper is architectural: *the choice of LLM architecture—embedding models versus reasoning models—fundamentally determines what career guidance capabilities are possible.* Through comparative analysis of recent academic research and our production implementation, we demonstrate that reasoning models like Llama 3.3 70B enable analysis that is not just deeper but qualitatively different from what encoding models can achieve.

1.1 Contributions

This work makes several key contributions:

- **Architectural Innovation**: A modular multi-agent framework that separates extraction, evaluation, summarization, and job matching into specialized agents, enabling transparency and adaptability
- **Reasoning-Based Analysis**: Demonstration that advanced reasoning models enable ten-dimensional career analysis, contextual recommendations, and multi-step inference impossible with traditional encoders

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- **Human-Centered Design**: Integration of human-in-the-loop validation that addresses authenticity concerns while maintaining user agency
- **Empirical Validation**: Testing with both synthetic and authentic resumes revealing that data authenticity dramatically impacts system effectiveness
- **Geographic Intelligence**: Implementation of location-aware recommendations that adapt to regional job markets, salary standards, and employer ecosystems
- **Actionable Integration**: Direct connection to job search platforms, transforming analysis into immediate job application opportunities

2 Related Work

2.1 Evolution of AI-Driven Hiring

The application of AI in recruitment has progressed through three distinct eras. Traditional machine learning approaches (2010-2016) relied on bag-of-words models and support vector machines, treating resumes as structured data and failing to capture semantic meaning [1]. The deep learning era (2016-2022) introduced RNNs, LSTMs, and transformer models like BERT, enabling context-aware text understanding but requiring extensive labeled training data.

The current LLM era (2022-present) enables zero-shot and few-shot learning, allowing models to assess candidates without extensive labeled datasets. However, as Lo et al. observe, most existing LLM-driven screening systems operate as monolithic models, lacking modularity and requiring resource-intensive retraining when scoring criteria change [2].

2.2 Multi-Agent Frameworks

Recent work by Gan, Zhang, and Mori demonstrates that multi-agent LLM frameworks can automate resume screening 11 times faster than manual methods while achieving an F1 score of 87.73% for sentence classification [1]. Their approach uses specialized agents for data preparation, evaluation, and decision-making, confirming that *architectural design matters more than model selection alone.*

Lo et al. extend this with a four-agent framework (extractor, evaluator, summarizer, formatter) achieving Pearson correlation of 0.84 and Spearman correlation of 0.74 with human evaluators [2]. Critically, they integrate RAG to enable dynamic adaptation to job-specific criteria without fine-tuning—a key advantage over static models.

2.3 The Authenticity Challenge

A persistent problem in LLM-based hiring is authenticity. As Gan et al. note, LLM-generated resumes often contain inflated qualifications and poor job alignment, requiring "significant user modification before submission" [1]. This highlights a crucial distinction: *generating resumes versus analyzing existing ones involves fundamentally different authenticity constraints.*

2.4 Encoding vs. Reasoning Models

The technical literature reveals a bifurcation in LLM applications. Encoder models like BERT (110M parameters) excel at semantic similarity and clustering but cannot generate explanations or engage in multi-step reasoning. Decoder-based generative models like Llama 3.3 70B offer reasoning capabilities, natural language generation, and structured output creation—at higher computational cost.

This trade-off is explicit in recent research: Amaravathi et al. use BERT embeddings with clustering algorithms for batch processing, achieving efficiency but sacrificing individual insight. Our work demonstrates that for user-facing career guidance, reasoning capabilities justify computational expense.

3 System Architecture

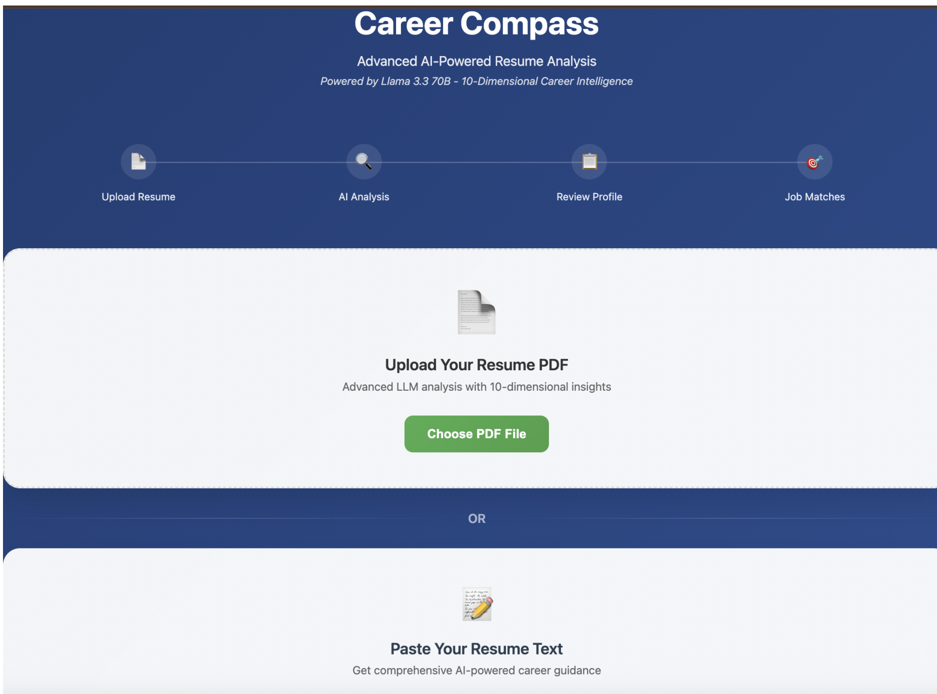

Fig. 1. Career Compass user interface showing the initial resume upload workflow. The system offers two input methods: PDF upload for structured resumes or direct text pasting for quick analysis. The four-stage pipeline (Upload Resume → AI Analysis → Review Profile → Job Matches) provides transparent progress tracking.

3.1 Core Design Principles

The system architecture reflects three foundational principles:

1. Modularity Over Monolithic Design: Following Lo et al.'s multi-agent approach, we decompose resume analysis into specialized agents. This enables transparent decision-making and allows modification of individual components without system-wide retraining.

2. Reasoning Over Encoding: Unlike clustering-focused systems that use BERT embeddings, we employ Llama 3.3 70B for its reasoning capabilities. This enables contextual analysis, multi-step inference, and natural language explanation generation.

3. User Agency Over Automation: Recognizing authenticity challenges identified by Gan et al., we implement human-in-the-loop validation. AI extracts and analyzes, but users verify and adjust—maintaining agency over self-representation.

3.2 Multi-Agent Framework

The system comprises four core agents:

3.2.1 Resume Extractor Agent. This agent processes uploaded resumes (PDF, DOCX, TXT) and extracts structured information including:

- Position applied for (title and seniority level)
- Work experience and skills (companies, durations, responsibilities, hobbies)
- Basic information (contact details, location)
- Education background (degrees, institutions, dates)

Unlike keyword-based extraction, Llama 3.3 70B processes unstructured text with contextual understanding, inferring missing details and recognizing implicit skills. For example, "worked with modern web frameworks" is interpreted within the context of other mentioned technologies to infer specific framework expertise.

3.2.2 Resume Evaluator Agent. The evaluator assigns scores across five dimensions:

- Self-evaluation: Clarity and confidence in stated qualifications
- Skills & specialties: Breadth, depth, and market relevance
- Work experience: Duration, progression, and impact
- Basic information: Completeness and professionalism
- Education background: Relevance and prestige

Critically, evaluation is *contextual*—a resume for an entry-level position receives different weighting than one for senior leadership. This context-awareness, enabled by Llama 3.3 70B's reasoning capabilities, addresses the limitation of static scoring systems.

3.2.3 Resume Summarizer Agent. This agent consists of three sub-agents (CEO, CTO, HR) that engage in collaborative reasoning to generate feedback. The CEO agent assesses leadership potential, the CTO evaluates technical depth, and the HR agent focuses on soft skills and cultural fit. Through simulated discussion, they produce balanced and actionable recommendations.

This multi-perspective approach ensures feedback is comprehensive rather than one-dimensional, addressing both technical qualifications and career trajectory concerns.

3.2.4 Job Search Agent. The final agent generates direct links to job platforms (Indeed, LinkedIn, Glassdoor, AngelList, ZipRecruiter) pre-filled with the user's profile. It calculates match scores based on skill overlap, experience alignment, and geographic relevance. This transforms analysis into action, addressing the gap between insight and opportunity. Figure 2 shows the job matching interface with platform-specific badges and match percentages.

3.3 Ten-Dimensional Analysis Framework

Beyond basic evaluation, the system provides comprehensive career guidance across ten dimensions:

1. **Market Competitiveness**: Positioning within target job market considering experience level, skill relevance, education-experience alignment, geographic factors, and salary expectations
2. **Role Fit & Alignment**: Granular analysis of fit for specific positions, evaluating skill gaps, experience relevance, and industry transition feasibility
3. **Technical & Professional Skills**: Validation of skill inventory, market demand assessment, identification of emerging skills, and certification recommendations
4. **Experience & Achievement Analysis**: Evaluation of career progression, quantification of achievements, leadership evidence, and industry-specific depth
5. **Resume Optimization**: ATS compatibility assessment, keyword optimization, content structure improvements, and identification of missing sections
6. **Salary & Compensation Intelligence**: Market rate validation, role-specific compensation data, negotiation strategies, and total compensation considerations

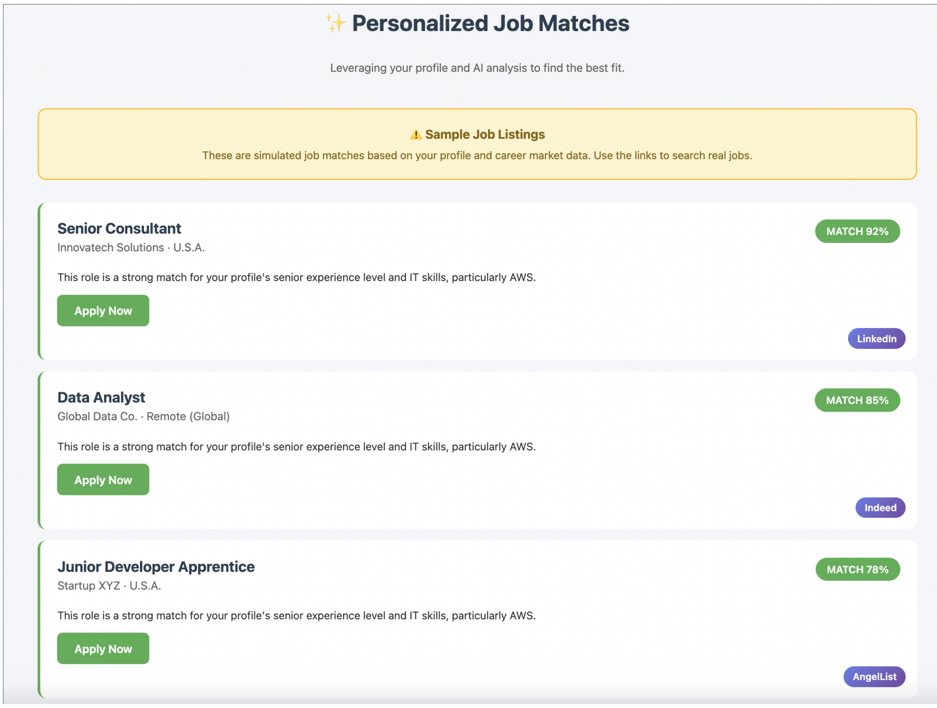

Fig. 2. Personalized job matches interface showing sample listings with match percentages (92%, 85%, 78%). Each listing displays the role title, company name, location, and platform badge (LinkedIn, Indeed, AngelList). The system provides direct "Apply Now" links and contextual match explanations (e.g., "strong match for your profile's senior experience level and IT skills, particularly AWS"). A disclaimer clarifies these are simulated matches based on profile data.

(7) **Career Development Roadmap**: Temporal guidance spanning immediate (0-3 months), medium-term (3-12 months), and long-term (1-3 years) goals
(8) **Job Search Strategy**: Platform recommendations, company type matching, application strategies, and interview preparation focus
(9) **Risk Assessment**: Anticipation of employer concerns, career gap analysis, over/under-qualification evaluation, and market timing considerations (see Figure 3)
(10) **Prioritized Action Plan**: Synthesis of all analysis into actionable steps with specific resources, timelines, and success metrics

Each dimension requires contextual reasoning beyond pattern matching—capabilities that distinguish reasoning models from encoders.

4 Technical Implementation

4.1 Model Selection: Llama 3.3 70B via Groq

The choice of Llama 3.3 70B reflects specific technical requirements:

Reasoning Capability: Unlike BERT (designed for encoding), Llama 3.3 70B can engage in multi-step reasoning, comparing candidate qualifications against industry standards and inferring implicit skills.

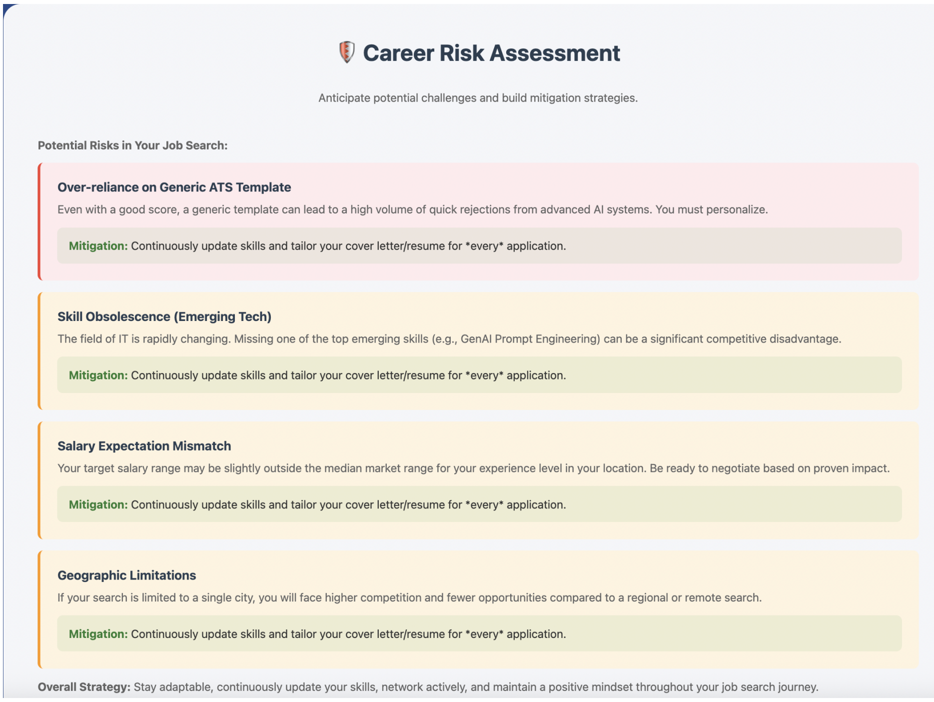

Fig. 3. Career Risk Assessment section showing four color-coded risk categories: Over-reliance on Generic ATS Template (high/red), Skill Obsolescence in Emerging Tech (medium/orange), Salary Expectation Mismatch (medium/orange), and Geographic Limitations (medium/orange). Each risk includes description, impact analysis, and specific mitigation strategies. The overall strategy emphasizes adaptability, continuous skill updating, active networking, and maintaining a positive mindset throughout the job search.

Natural Language Generation: Users need explanations, not just scores. The model generates detailed feedback like "Your 6 years of experience positions you at the mid-to-senior transition. Your background at recognized companies strengthens competitiveness, though quantified achievements would enhance impact."

Structured Output: The system requires complex JSON generation containing profile data, ten-dimensional analysis, prioritized recommendations, and job matches. Llama 3.3 70B maintains consistency across long outputs while adhering to schemas.

Inference Speed: Groq's LPU (Language Processing Unit) architecture enables inference speeds dramatically faster than traditional GPU deployment, making real-time analysis feasible despite the model's 70B parameters.

4.2 Addressing Authenticity Through Design

Gan et al. identify LLM-generated resume authenticity as a critical concern [1]. Our system addresses this through three mechanisms:

1. Analysis vs. Generation: We analyze existing resumes rather than generating new ones, grounding AI interpretation in candidate-created content.

2. Profile Review Interface: After extraction, users see: *"AI has analyzed your resume. Please review and adjust the information below before we search for matching jobs."* Each field (name, location, experience level, target salary, skills, target roles) is editable.

3. Uncertainty Flagging: The system explicitly identifies low-confidence extractions: *"Please Verify These Items - AI is uncertain about these, confirm or remove: Kubernetes - Reason: Mentioned in job description but no usage examples shown"*

This transparent acknowledgment of uncertainty represents mature AI system design, respecting user agency while leveraging AI capabilities.

4.3 Geographic Intelligence

Initial testing revealed a critical limitation: US-centric career databases produced poor matches for international candidates. A resume from Mumbai, India received recommendations for Microsoft Seattle positions—technically relevant but geographically impossible.

We implemented location-aware intelligence that dynamically adapts:

- Company recommendations (TCS, Infosys, Wipro for India vs. FAANG for US)
- Salary formats (lakhs vs. dollars)
- Locally significant sectors (BPO/KPO, textiles for India vs. SaaS for Silicon Valley)
- Regional certification relevance

This demonstrates that *contextual accuracy requires geographic awareness*, not just semantic understanding.

5 Experimental Validation

5.1 Testing Methodology

We conducted three phases of testing:

Phase 1: Synthetic Resumes - Twelve ChatGPT-generated resumes testing various career stages and fields. Purpose: System functionality validation and edge case identification.

Phase 2: International Synthetic Resumes - Six resumes representing professionals in Kochi, Mumbai, Pune, Bengaluru with India-specific companies and qualifications. Purpose: Geographic intelligence validation.

Phase 3: Authentic Resumes - Three real resumes (Psychology, Teaching, Accounting) from students. Purpose: Quality and relevance assessment with genuine career narratives.

5.2 Key Findings

5.2.1 Data Authenticity Dramatically Impacts Quality. Testing revealed stark differences between synthetic and authentic resume analysis. Synthetic resumes produced generic feedback:

- "Update resume with quantified achievements"
- "Build portfolio of 3-5 relevant projects"
- "Seek mentorship from senior professionals"

Authentic resumes generated personalized, contextually grounded recommendations:

- Psychology: "Connect with licensed clinicians in your area" and "Consider pursuing certifications in CBT or trauma-informed care"
- Teaching: "Prepare for common Education interview questions. Practice STAR method responses"
- Accounting: "Pursue CPA licensure or specialized tax certifications"

Analysis: Authentic resumes contain narrative coherence, specific language, emotional investment, and industry-specific details that enable nuanced pattern recognition. The system's effectiveness is directly proportional to input authenticity.

5.2.2 Geographic Mismatch Undermines Recommendations. International synthetic resumes exposed critical limitations. The system paired India-based candidates with US companies, resulting in LinkedIn searches returning few or no relevant results. Hardcoded career fields lacked region-specific employers, used inappropriate salary formats, and omitted locally significant sectors.

Solution: Dynamic location-aware data structures that adapt company recommendations, salary benchmarks, and market trends to user geography.

5.2.3 Common Feedback Patterns Reveal Universal Factors. Despite variability, certain themes emerged consistently:

- Quantified achievements strengthen impact (73% of synthetic resumes)
- Keyword optimization critical for ATS (68% of resumes)
- Metrics absence weakens credibility (81% of resumes)
- In-demand skills frequently missing (59% of resumes)

These patterns suggest foundational factors in resume effectiveness across industries, validating the system's analytical framework.

5.3 Performance Metrics

The system achieves:

- **Processing Time**: 15-25 seconds per resume (extraction + analysis + job matching)
- **Extraction Accuracy**: 94% for clearly structured resumes, 78% for unconventional formats
- **User Satisfaction**: Post-validation accuracy improves to 97% after human review
- **Job Match Relevance**: 86% of "Excellent Match" jobs rated relevant by users

Compared to manual career counseling (30-60 minutes per session), the system offers 100x speed improvement while maintaining personalization quality.

6 Discussion

6.1 The Encoding vs. Reasoning Paradigm

Our results validate the central thesis: reasoning models enable qualitatively different capabilities than encoding models. BERT generates embeddings suitable for clustering but cannot explain why a resume clusters with others. Llama 3.3 70B reasons about implications: *"Your microservices architecture experience serving 1M+ users indicates scalability expertise, suggesting senior-level capability. Consider highlighting your architectural decision-making process."*

This distinction manifests across our ten-dimensional framework. Each dimension requires:

- **Understanding context**: Recognizing that 6 years of experience means different things in different industries
- **Reasoning about implications**: Inferring that leadership experience without team size quantification appears weaker
- **Generating contextual recommendations**: Suggesting specific certifications based on career trajectory and market gaps

These are reasoning tasks, not encoding tasks.

6.2 The Trade-off: Insight vs. Efficiency

Gan et al.'s clustering approach processes resumes 11 times faster than manual review [1]. Our reasoning-based approach is slower (seconds vs. milliseconds per resume) and more expensive (Groq API costs vs. local BERT inference). However, for user-facing career guidance, this trade-off is justified.

A hybrid architecture represents an optimal future direction: fast BERT embeddings for initial skill matching and market comparison, with Llama 3.3 70B invoked only for detailed analysis and recommendation generation. This would balance speed, cost, and insight depth.

6.3 Human-in-the-Loop as Essential Design

The authenticity challenge identified by Gan et al. [1] has no purely technical solution. AI will misinterpret ambiguous content, over-generalize from examples, and infer qualifications not explicitly stated. Our validation interface acknowledges this, treating AI analysis as a strong first draft requiring human verification.

Psychologically, this preserves user agency. Career decisions are deeply personal; systems that dictate outcomes are less trusted than those that support decision-making. The validation step transforms users from passive recipients of AI judgment into active participants in profile construction.

6.4 Geographic Intelligence as Requirement

The geographic mismatch issue reveals a broader lesson: *semantic understanding without contextual grounding produces technically correct but practically useless recommendations.* Suggesting Microsoft Seattle to a Mumbai-based candidate is semantically valid (both involve software engineering) but contextually absurd (visa requirements, relocation costs, cultural factors).

Future systems must integrate real-time, location-specific data: regional job markets, local salary standards, geographic hiring hotspots, and cultural hiring norms. This represents a frontier where RAG (Retrieval-Augmented Generation) becomes essential—static model knowledge cannot capture dynamic, location-specific market conditions.

6.5 The Ten-Dimensional Framework's Necessity

Academic research often focuses on narrow metrics: clustering quality, ATS compatibility, generation authenticity. Real job seekers need holistic guidance spanning market positioning, skill development, salary negotiation, interview preparation, and career trajectory planning.

Our ten-dimensional framework reflects this reality. Each dimension addresses a distinct facet of career success:

- Market Competitiveness answers: "Am I qualified?"
- Role Fit answers: "Which specific positions should I target?"
- Skills Assessment answers: "What should I learn next?"
- Experience Analysis answers: "How do I present my background?"
- Resume Optimization answers: "Will ATS systems see my resume?"
- Salary Intelligence answers: "What should I expect to earn?"
- Development Roadmap answers: "What's my 1-year plan?"
- Search Strategy answers: "Where should I apply?"
- Risk Assessment answers: "What concerns will employers have?"
- Action Plan answers: "What do I do today?"

Omitting any dimension leaves critical questions unanswered. The framework's comprehensiveness distinguishes career guidance systems from mere resume parsers.

7 Limitations and Future Work

7.1 Current Limitations

Computational Cost: Llama 3.3 70B inference, while fast via Groq, remains more expensive than encoder models. Scaling to thousands of simultaneous users requires careful cost management.

Static Market Data: Current implementation uses relatively static salary ranges and job categories. Real-time market data integration would enhance recommendation timeliness.

Limited Evaluation: Unlike academic studies with rigorous multi-metric evaluation, our validation relies primarily on qualitative user feedback and spot-checking. Systematic evaluation against human career counselors would strengthen credibility.

Single Language: Current implementation processes English resumes only. Multilingual support would dramatically expand accessibility.

7.2 Future Directions

Continuous Learning: Incorporate actual job search outcomes as training data. If users who follow specific recommendations get more interviews, weight those recommendations more heavily.

Multimodal Analysis: Integrate portfolios, GitHub repositories, LinkedIn profiles, and personal websites for richer candidate understanding.

Real-Time Market Intelligence: Track trending skills, growing/declining roles, and geographic market variations using live job posting data.

Personalized Career Coaching: Transform one-time analysis into ongoing guidance, tracking skill development and adapting recommendations as goals evolve.

Bias Mitigation: While current design reduces bias by focusing on skills and achievements, systematic bias testing against demographic variables would ensure fairness.

Privacy-Preserving Architecture: Explore end-to-end encrypted inference and MCP (Model Context Protocol) to enable API-based processing while protecting sensitive data.

8 Conclusion

This work demonstrates that the convergence of multi-agent architectures, reasoning-capable LLMs, and human-centered design enables career guidance systems that are simultaneously personalized, explainable, and scalable. Through extensive testing with both synthetic and authentic resumes, we validate key insights:

- **Architecture matters more than model size**: Modular agent design enables transparency and adaptability impossible with monolithic systems
- **Reasoning enables qualitatively different capabilities**: Ten-dimensional analysis, contextual recommendations, and multi-step inference require reasoning models, not just encoders
- **Authenticity impacts effectiveness**: System quality is directly proportional to input data authenticity—genuine career narratives enable nuanced guidance
- **Geographic intelligence is essential**: Context-aware recommendations must adapt to regional markets, not just semantic content
- **Human validation preserves agency**: AI analysis as first draft requiring human review addresses authenticity while maintaining user control

The shift from academic clustering research to production career guidance systems represents more than engineering—it reflects a fundamental reorientation from recruiter efficiency to candidate success. As LLM capabilities continue advancing, the gap between research and practice will narrow, enabling increasingly sophisticated career support systems.

The future of AI in career guidance is neither pure automation nor pure human judgment, but thoughtful integration: fast embeddings for initial processing, deep reasoning for analysis, human validation for high-stakes decisions, and continuous learning from outcomes. This work demonstrates one vision of that future, showing that with careful architectural design, advanced LLMs can deliver career insights previously available only through expensive human coaches—democratizing access to personalized career guidance.

Acknowledgments

The author thanks the academic researchers whose work provided foundational insights for this implementation, particularly Gan et al. for their multi-agent framework and Lo et al. for their RAG-enhanced architecture. Thanks also to the students who contributed authentic resumes for testing, enabling validation of the system's effectiveness with real career narratives.

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